RESEARCH / WHITEPAPER

Environmental Impact Analysis

Carbon footprint and energy efficiency of local, offline video processing



LOCAL COMPUTATION.

A CLOSER LOOK AT THE FOOTPRINT.

01 Executive summary

Videobeaux is a local, offline video utility. **AI is used in only a few program modes.** Most modes use non-AI processing methods and software libraries. Both AI and non-AI modes run on the user's own hardware, without cloud processing, hosted services, or remote inference.

This whitepaper examines the environmental implications of that architecture. In the modes that use AI, inference runs locally rather than in a remote data center. Across the software, local processing avoids the need to upload media to a service and retrieve remotely processed results.

02 Localized vs. cloud AI

For the few modes that use AI, the distinction is between remote cloud computation and local inference:

CLOUD-SCALE GENERATIVE AI

Cloud-hosted AI uses remote computing infrastructure, with energy required for servers, supporting systems such as cooling, and data transmission. Its footprint varies with the workload, hardware efficiency, and electricity supply. Local processing removes the need for a remote processing service; it does not, by itself, prove lower total emissions. [1, 2]

● VIDEOBEAUX / LOCAL UTILITY AI

Only a few Videobeaux modes use specialized AI models for functional tasks such as Text-to-Speech, Transcription, and Background Removal. These processes run on the user's machine without a remote inference service. Most other modes use non-AI methods and software libraries, with the same local, offline approach. Active processing draws power from the user's own hardware.

03 Recorded energy & emissions

CodeCarbon 3.3.1 telemetry captured energy and estimated emissions during local use on Apple M5 Max hardware with 128 GB RAM. The two substantive records are summarized below. Tracking covered the machine; three near-empty records were excluded.

REPORTED METRIC	RECORD A	RECORD B
Energy consumed (kWh)	**0.002188**	**0.004089**
Estimated emissions (g CO2e)	**0.815**	**1.523**
Mean CPU power (W)	**16.13**	**24.62**
Mean RAM power (W)	**11.40**	**11.40**
Reported GPU power (W)	**0.00**	**0.00**

Internal telemetry, October 4, 2026. Values rounded; emissions converted from kilograms to grams. Records A and B identify the two observations summarized here.

04 Four-hour usage scenario

0.112-0.147
kWh / EXTRAPOLATED

41.8-54.8
g CO2e / EXTRAPOLATED

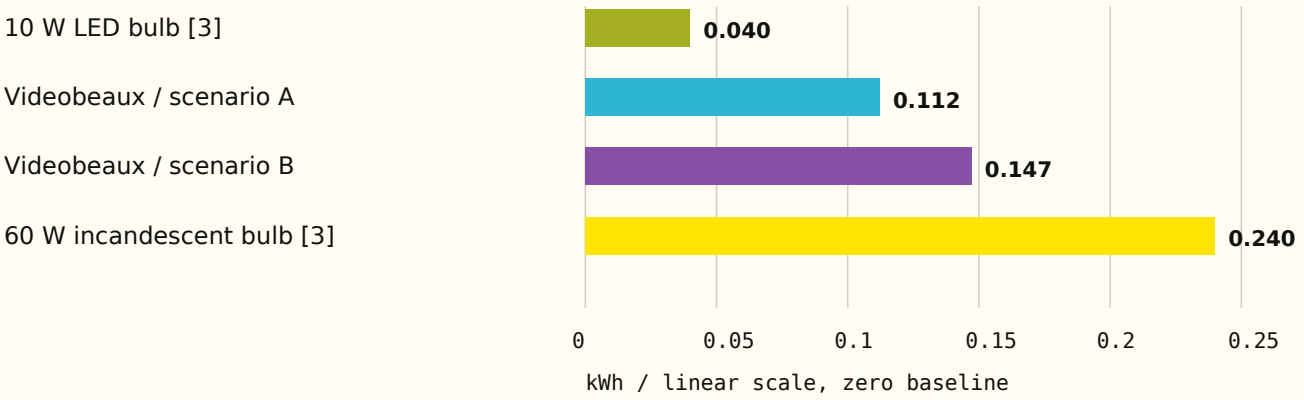
These are **four-hour extrapolations, not measured four-hour results.** Each record is scaled independently: recorded total divided by its logged elapsed time, then multiplied by four hours. The range represents the two scenarios, not a statistical confidence interval. It assumes unchanged workload and electricity carbon intensity.

Measurement boundary. CodeCarbon estimates tracked compute energy and associated emissions [1]. Machine-level figures can include other activity; no idle baseline was supplied. A GPU field of zero does not establish zero GPU use or energy. These are not wall-outlet readings, a full lifecycle footprint, or a cloud comparison.

05 Everyday context

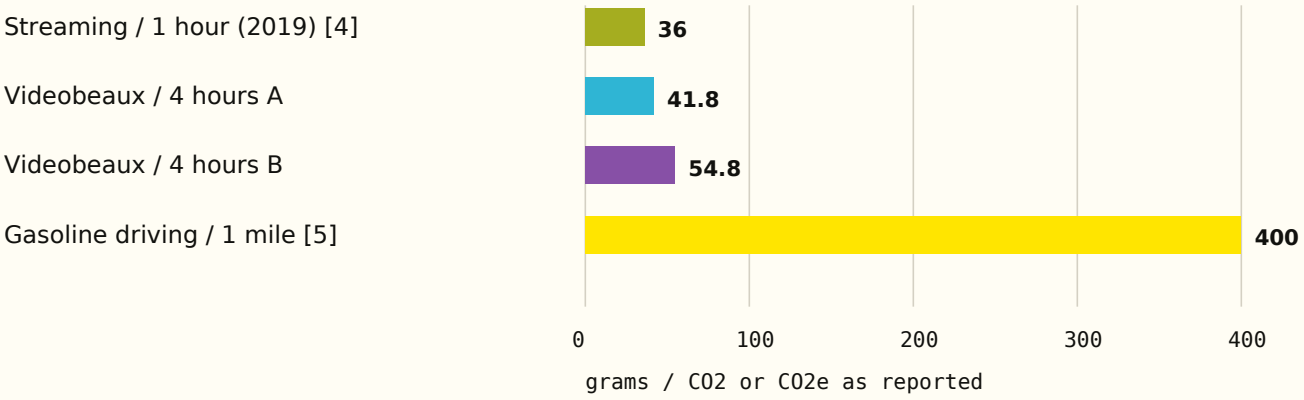
Everyday lighting, watching video, and driving offer familiar reference points. These comparisons show scale, not equivalent work or a full lifecycle ranking. Videobeaux values remain extrapolated machine-level estimates.

ENERGY / FOUR HOURS



Lighting values are calculated as watts x 4 hours / 1,000. The 10 W LED is an explicit example within the DOE 5.9-10.5 W range for an 800-lumen LED replacing a 60 W incandescent. The software scenarios use 2.8-3.7 times the LED energy, or 47-61% of the incandescent energy. [3]

EMISSIONS / SELECTED ACTIVITIES



The software estimates are numerically about 1.2-1.5 times the IEA 2019 global streaming estimate, and 10-14% of the EPA one-mile gasoline-car tailpipe CO2 estimate. Streaming and driving sources report CO2; CodeCarbon reports CO2-equivalents. These have different boundaries and energy assumptions. [4, 5]

06 The footprint of development

Human-written software also has a carbon footprint. Writing, debugging, compiling, testing, and packaging software require computing resources. Development tools, build pipelines, storage, and networking can contribute electricity-related emissions. Manufacturing the computers used for this work adds embodied emissions. [2]

A useful development assessment separates **operational emissions** from electricity use and **embodied emissions** allocated from hardware manufacture and disposal. Energy use multiplied by electricity carbon intensity estimates operational emissions; allocating hardware emissions requires lifecycle and usage information. [2]

For Videobeaux, reusing existing hardware, avoiding unnecessary builds and repeated processing, and choosing efficient tools are practical opportunities to reduce resource use. The telemetry does not measure development work, so no numerical development footprint is claimed here.

07 Efficiency by design

Videobeaux keeps media processing local and offline, without hosted processing services. Most modes use non-AI methods and software libraries; only a few use specialized AI models.

The recorded telemetry provides a limited view of operational energy use on one machine. A fuller assessment would include representative tasks, completed-output counts, an idle baseline, hardware lifecycle impacts, and a matched cloud workload before claiming comparative savings.

LOCAL. OFFLINE. SERVICELESS.

AI IN A FEW MODES. NON-AI IN MOST.

REFERENCES / CLICKABLE SOURCES

[1] CodeCarbon 3.3 methodology (docs.codecarbon.io)
[2] Green Software Foundation, SCI specification (sci.greensoftware.foundation)
[3] DOE, Purchasing Energy-Efficient Light Bulbs (energy.gov)
[4] IEA, Streaming video footprint: 2019 estimate, updated 2020 (iea.org)
[5] EPA, Greenhouse Gas Emissions from a Typical Passenger Vehicle (epa.gov)